

ASSIGNMENT #8 10/27/22

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SECTION 5.1

Exercise (5.1.10 — Noah Palmer). Let p be a prime. Let A and B be two cyclic groups of order p with generators x and y , respectively. Set $E = A \times B$ so that E is the elementary abelian group of order $p^2 : E_{p^2}$. Prove that the distinct subgroups of E of order p are

$$\langle x \rangle, \quad \langle xy \rangle, \quad \dots, \quad \langle xy^{p-1} \rangle, \quad \langle y \rangle$$

Proof. Let p be a prime, $\langle x \rangle = A$ and $\langle y \rangle = B$ be two cyclic groups of order p , and set $E = A \times B$ so that E is the elementary abelian group of order p^2 . Since A and B are cyclic groups of prime order it follows that each element of A and B have order p . Hence, each of the subgroups

$$\langle x \rangle, \quad \langle xy \rangle, \quad \dots, \quad \langle xy^{p-1} \rangle, \quad \langle y \rangle$$

will have order p . Additionally, each of these groups are unique since each y^i is unique within its own cyclic group which implies that the subgroups generated by $\langle (1, 1, \dots, x, \dots, y^i, \dots 1) \rangle$ will be unique for each $1 \leq i < p$. From example 3 in section 5.1 of Dummit and Foote the number of subgroups of order p in an elementary abelian group of order p^2 is $p + 1$. Since, we have given a list of exactly $p + 1$ distinct groups, we're done.

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SECTION 5.2

Exercise (5.2.4 — Noah Palmer). In each of parts (a) to (d) determine which pairs of abelian groups listed are isomorphic.

- (a) $\{4, 9\}$, $\{6, 6\}$, $\{8, 3\}$, $\{9, 4\}$, $\{6, 4\}$, $\{64\}$.
- (b) $\{2^2, 2 \cdot 3^2\}$, $\{2^2 \cdot 3, 2 \cdot 3\}$, $\{2^3 \cdot 3^2\}$, $\{2^2 \cdot 3^2, 2\}$.

- (c) $\{5^2 \cdot 7^2, 3^2 \cdot 5 \cdot 7\}, \{3^2 \cdot 5^2 \cdot 7, 5 \cdot 7^2\}, \{3 \cdot 5^2, 7^2, 3 \cdot 5 \cdot 7\}, \{5^2 \cdot 7, 3^2 \cdot 5, 7^2\}.$
 (d) $\{2^2 \cdot 5 \cdot 7, 2^3 \cdot 5^3, 2 \cdot 5^2\}, \{2^3 \cdot 5^3 \cdot 7, 2^3 \cdot 5^3\}, \{2^2, 2 \cdot 7, 2^3, 5^3, 5^3\}, \{2 \cdot 5^3, 2^2 \cdot 5^3, 2^3, 7\}.$

Proof. We use the fact that $Z_m \times Z_n \cong Z_{mn}$ if and only if $\gcd(m, n) = 1$ to rearrange the abelian groups into isomorphic pairs, where possible.

- (a) Note that $Z_4 \times Z_9 \cong Z_9 \times Z_4$. Thus, $\{4, 9\} \cong \{9, 4\}.$
 (b) Note that $Z_{2^2} \times Z_{2 \cdot 3^2} \cong Z_{2^2 \cdot 3^2} \times Z_2$. Thus, $\{2^2, 2 \cdot 3^2\} \cong \{2^2 \cdot 3^2, 2\}.$
 (c) Note that $Z_{5^2 \cdot 7^2} \times Z_{3^2 \cdot 5 \cdot 7} \cong Z_{3^2 \cdot 5^2 \cdot 7} \times Z_{5 \cdot 7^2}$. Also, $Z_{3^2 \cdot 5^2 \cdot 7} \times Z_{5 \cdot 7^2} \cong Z_{5^2 \cdot 7} \times Z_{3^2 \cdot 5} \times Z_{7^2}$.
 Thus, $\{5^2 \cdot 7^2, 3^2 \cdot 5 \cdot 7\} \cong \{3^2 \cdot 5^2 \cdot 7, 5 \cdot 7^2\} \cong \{5^2 \cdot 7, 3^2 \cdot 5, 7^2\}.$
 (d) Note that $Z_{2^2} \times Z_{2 \cdot 7} \times Z_{2^3} \times Z_{5^3} \times Z_{5^3} \cong Z_{2^2 \cdot 5^3} \times Z_{2 \cdot 7 \cdot 5^3} \times Z_{2^3} \cong Z_{2 \cdot 5^3} \times Z_{2^2 \cdot 5^3} \times Z_{2^3} \times Z_7$.
 Therefore, $\{2^2, 2 \cdot 7, 2^3, 5^3, 5^3\} \cong \{2 \cdot 5^3, 2^2 \cdot 5^3, 2^3, 7\}.$

□

Exercise (5.2.9 — Noah Palmer). Let $A = Z_{60} \times Z_{45} \times Z_{12} \times Z_{36}$. Find the number of elements of order 2 and the number of subgroups of index 2 in A .

Proof. Begin by rewriting the direct product in its invariant form $Z_{60} \times Z_{45} \times Z_{12} \times Z_{36} \cong Z_{180} \times Z_{60} \times Z_3 \times Z_{36} \cong Z_{180} \times Z_{180} \times Z_{12} \times Z_3$. Now, note that each of Z_{180} and Z_{12} has an element of order 2 whereas Z_3 does not. Hence, there will be 2^3 elements of order less than or equal to 2. From these we must exclude the identity since it has order 1. Thus, there are 7 elements of order 2.

To find the subgroups of index 2 in A consider the elementary divisor factorization of $A = Z_5 \times Z_5 \times Z_4 \times Z_4 \times Z_4 \times Z_9 \times Z_9 \times Z_3 \times Z_3$. Now, any subgroup of order 583,200 will have index 2 in A . From the above factorization, and the prime factorization of 583,200, these subgroups will have the form $A' \times Z_5 \times Z_5 \times Z_9 \times Z_9 \times Z_3 \times Z_3$ where A' is a subgroup of $Z_4 \times Z_4 \times Z_4$ with order 32. There are three ways to select such a subgroup (e.g. $Z_4 \times Z_4 \times Z_2$, $Z_2 \times Z_4 \times Z_4$, and $Z_4 \times Z_2 \times Z_4$). Thus, there are three subgroups of index 2 in A .

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